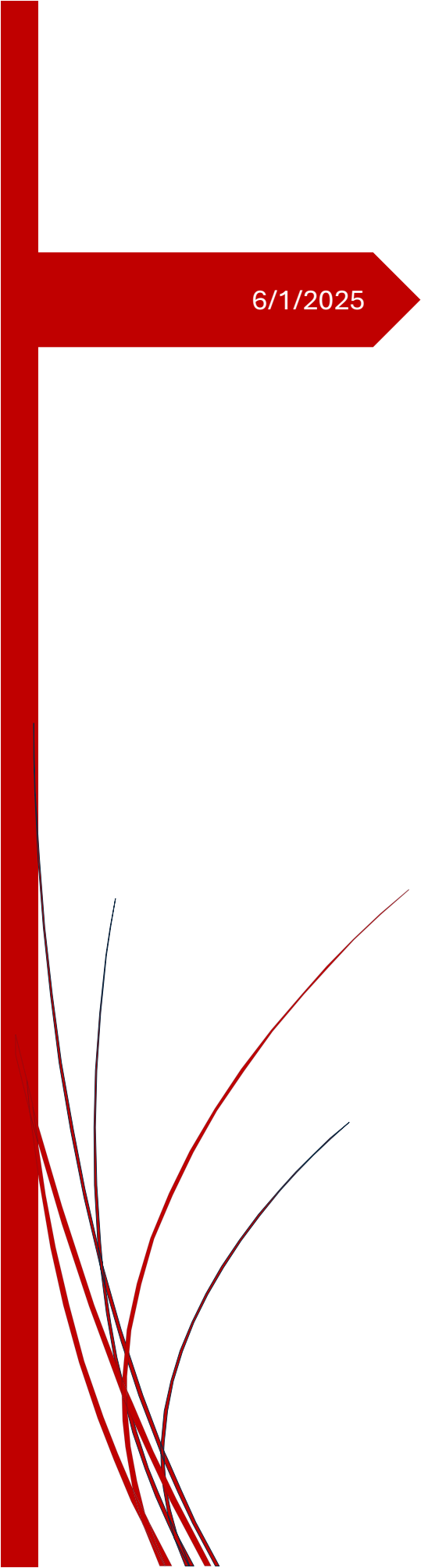
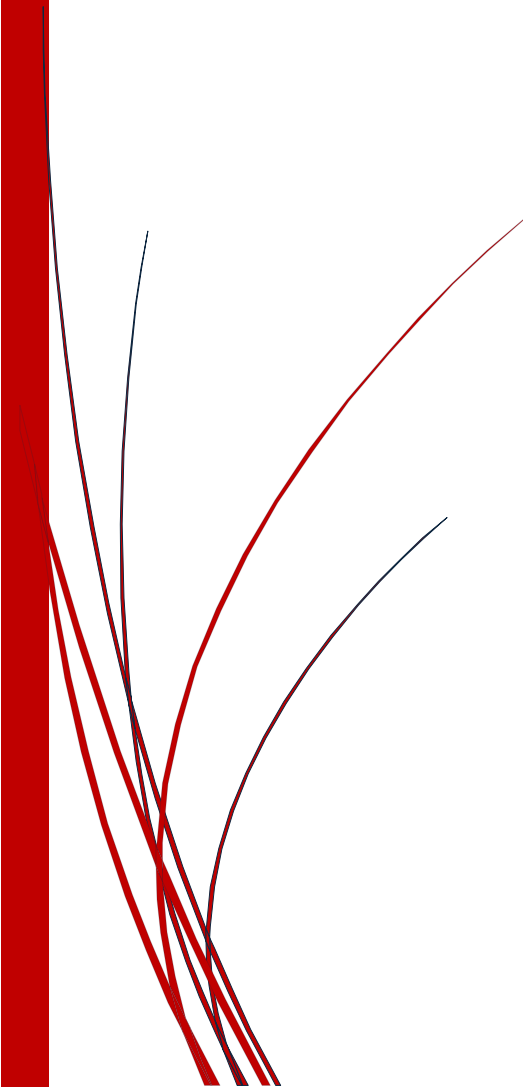


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6/1/2025

Database Standards

By : Daniyal Anwar

Abstract graphic consisting of several thin, curved lines in red and black, originating from the bottom left and extending upwards and to the right.

Daniyal Anwar

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Daniyal Anwar

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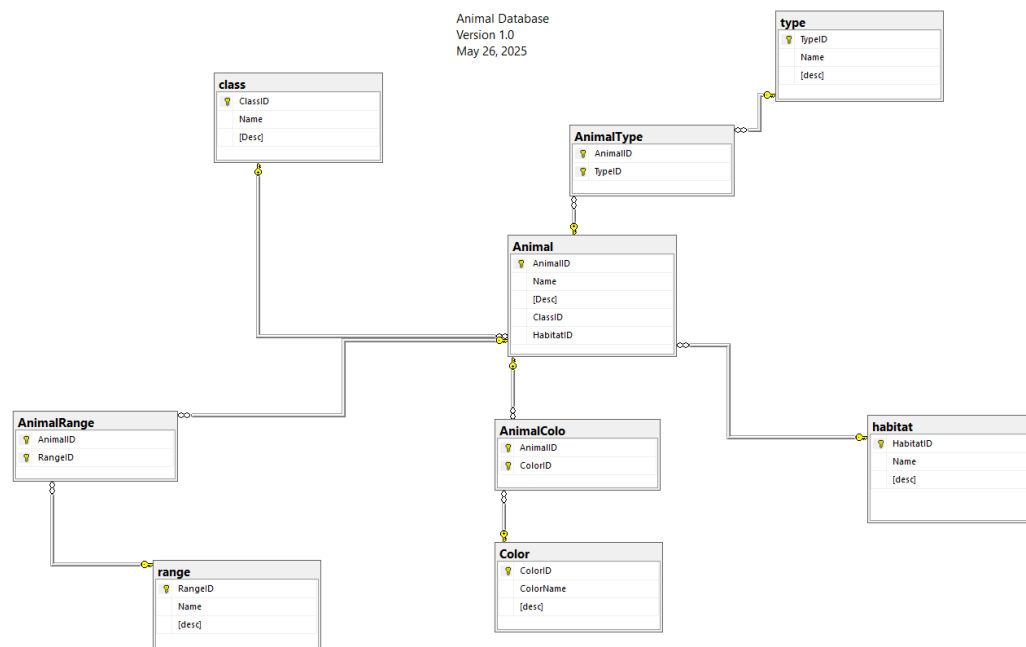
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SQL DATABASE STANDARD

Project Overview

The SQL Animal Database is designed to manage information about animals, their classifications, habitats, colors, and distribution ranges. It facilitates data retrieval for zoological studies, conservation efforts, and education.

Database Design Diagram



Data Dictionary

Animal Table

Field Name	Data Type	Nullable	Description	Example
AnimalID	Integer	No	Unique animal identifier	1
Name	Nvarchar(50)	No	Name of the animal	Lion
Desc	Nvarchar(MAX)	Yes	Description of the animal	Big Cat
ClassID	Integer	No	Foreign key to class table	101
HabitatID	Integer	No	Foreign key to Habitat table	102

Class Table

Field Name	Data Type	Nullable	Description	Example
ClassID	Integer	No	Unique class identifier	201
Name	Nvarchar(50)	No	Name of the animal class	Mammal
Desc	Nvarchar(MAX)	Yes	Description of class	Warm blooded

Habitat table

Field Name	Data Type	Nullable	Description	Example
HabitatID	Integer	No	Unique habitat identifier	206
Name	Nvarchar(50)	No	Name of habitat	Savanna
Desc	Nvarchar(MAX)	Yes	Description of habitat	Grassland

Type Table

Field Name	Data Type	Nullable	Description	Example
TypeID	Integer	No	Unique type identifier	302
Name	Nvarchar(50)	No	Name of type	Wild
Desc	Nvarchar(MAX)	Yes	Description of type	Living in natural condition

Animal Type Table (bridge table)

Field Name	Data Type	Nullable	Description	Example
AnimalID	Integer	No	Foreign key to Animal table	1
TypeID	Integer	No	Foreign key to Type table	103

Color Table

Field Name	Data Type	Nullable	Description	Example
ColorID	Integer	No	Unique Color identifier	305
ColorName	Nvarchar(50)	No	Name of Color	Golden
Desc	Nvarchar(MAX)	Yes	Description of color	Light, warm yellow

Animal Color Table (bridge table)

Field Name	Data Type	Nullable	Description	Example
AnimalID	Integer	No	Foreign key to Animal table	2
ColorID	Integer	No	Foreign key to Color table	106

Range Table

Field Name	Data Type	Nullable	Description	Example
RangeID	Integer	No	Unique Range identifier	62
Name	Nvarchar(50)	No	Name of geographical range	Africa
Desc	Nvarchar(MAX)	Yes	Description of geographical range	Sub-Saharan region

Animal Range Table (bridge table)

Field Name	Data Type	Nullable1	Description	Example
AnimalID	Integer	No	Foreign key to Animal table	6
RangeID	Integer	No	Foreign key to Range table	108

Standards Document

<i>Type of standard</i>	<i>Description</i>
Vendor Standards	Use Microsoft SQL Server as the primary database management system.
Design Standards	Ensure referential integrity through: • Primary keys • Foreign keys • Indexes • Database triggers
Database Naming Standards	Names should follow CamelCase convention (capitalize the first letter of each word). Do not use underscores (_) or spaces. Common abbreviations: ID: Identification Number Desc: Description
Datatype Standards	Numeric: Integer, Decimal

	Alphanumeric: Nvarchar, Nchar Do not use Binary Large Object (BLOB) datatypes.
Security Standards	Use Windows Authentication rather than SQL Server Authentication. Use of stored procedures and views, so that applications don't have full access to the databases. No use of embedded SQL in applications because of security risks.
Software Standards	Variables must include datatype prefixes for clarity: • iAnimalID for integer values • strAnimalName for string values Data manipulation routines must be developed and managed by database administrators (DBAs).
Change Control Standards	Utilize a staging environment for all database development and updates. Maintain a detailed change log recording: • Who performed changes • What changes were made • Why changes were necessary • When changes occurred

Summary of Assignment

Creating this SQL Animal Database has enhanced my understanding of database standards, relational database design, and practices ensuring data integrity and security. It reinforced the significance of thorough documentation and disciplined methodologies for managing structured data effectively.